

Hans-Peter Marshall

Associate Professor

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Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder, Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington, Thesis: *Wet snow densification*

Selected Professional Experience

- 08/2013-present Associate Professor, Department of Geosciences and Director, Cryosphere Geophysics And Remote Sensing (CryoGARS), Boise State University
- 07/2007-present Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers
Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire

Selected Honors and Awards

- 2020 Project Scientist, NASA SnowEx 2020 Mission
- 2017 NASA Robert H. Goddard Team Award for SnowEx Year 1
- 2012 Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
- 2010-13 New Investigator Program Award, NASA, 1st in Idaho history
- 2010 Young Scientist Award, Cryosphere Section, American Geophysical Union

Selected Related Refereed Publications [h-index=23, i10-index=40, citations=1525]

McGrath, D., Webb, R., Shean, D., Bonnell, R., **H.P. Marshall**, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during nasa's 2017 snowex campaign: Comparison to in situ, airborne, and satellite observations. *Water Resources Research*, 55(11):10026–10036

Lievens, H., Demuzere, M., **H.P. Marshall**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Giroto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12

Lund, J., Forster, R. R., Rupper, S. B., **H.P. Marshall**, Deeb, E. J., and Hashmi, M. Z. U. R. (2019). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. *Frontiers in Earth Science*, 7:318

Hedrick, A., Marks, D., Kormos, P., **H.P. Marshall**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063

Marshall, H.P., Hawley, R., and Tedesco, M. (2015). *Field measurements for remote sensing of the cryosphere*, chapter 14, pages 345–381. Remote Sensing of the Cryosphere. John Wiley & Sons, Ltd, ed. M. Tedesco